

ONLINE

NEWS

POPULARITY

AN6003 Analytics Strategy

Team Project - Report

Group Number 3

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Executive Summary

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1. Online news popularity and article promotion

1.1 Introduction and business objective

In this project, we take the role of Mashable's business analytics team in a hypothetical business scenario. The company aims to expand article reach and create breakout stories, which may support advertising revenue and brand awareness. Limited digital promotion resources mean that editors must decide which articles deserve priority support.

We propose ranking articles by their predicted sharing potential before publication. Waiting for substantial shares to accumulate may miss a timely opportunity to promote an article. Our business question is: How can Mashable identify articles with strong sharing potential and prioritise them for limited promotional support? Editors would review the shortlist alongside considerations such as relevance and editorial quality.

The analytical objective is to test whether Logistic Regression, Classification and Regression Trees (CART), or Random Forest identifies more popular articles within a fixed shortlist than random selection or historical channel ranking. The report develops this question through data preparation, modelling, evaluation and business recommendations.

1.2 Case context and data source

The Online News Popularity dataset links Mashable article characteristics with observed sharing outcomes [1]. The supplied raw CSV contains 39,644 articles published in 2013–2014, before cleaning. Earlier research explored popularity prediction before publication and changes to article features [2]. Our project focuses on selecting articles for limited promotional opportunities. Detailed cleaning decisions and the final sample are covered in the data preparation section.

1.3 Prediction target and study design

We define popular = 1 when recorded shares exceed 1,400 and popular = 0 when shares are 1,400 or fewer. This team convention places exactly 1,400 shares in the non-popular class, unlike the original dataset notes. Each model estimates the probability of exceeding this threshold; it does not predict total shares or extreme viral success.

All methods will use the team's shared chronological partitions. Training data support model fitting, validation data support method selection, and the final test set assesses the frozen choice. Historical shares provide the outcome for learning and evaluation and must never enter the predictors. Candidate inputs include article length, images, channel, planned publication day and sentiment, subject to verification that they are available before publication.

1.4 Selection rule and evaluation

Each method ranks the same evaluation articles and selects the highest-scoring 20%, rounding the number selected down to a whole article. Selection is global, with no separate channel or daily quotas. The 20% capacity is a project assumption; it determines shortlist size, while the 1,400-share threshold defines popularity.

We will select the approach primarily using validation Precision@20%: the number of popular articles selected divided by the total selected. Higher precision means more popular articles identified with the same capacity. Lift divides this precision by the popular proportion in the same evaluation set, showing the gain over random selection. Recommendations will consider whether improvement justifies additional model complexity.

1.5 Simple selection benchmarks

Random selection chooses the same number of articles uniformly without replacement. Its expected shortlist precision equals the popular proportion in that evaluation set. We use this mathematical expectation as the benchmark; an individual random draw may perform above or below it.

Historical channel ranking assigns each article its channel's training popular rate: popular training articles in that channel divided by all training articles in the channel. Articles are then ranked globally by this score. This tests whether the models improve on a simple rule based on topic category. Rates use training labels only, and a channel absent from training receives the overall training popular rate.

Equal scores use a fixed random order linked to article identifiers and shared across methods, without consulting the true sharing outcomes. Both benchmarks are selection methods for comparison, not rules for filtering training data or descriptions of Mashable's existing practice. R scripts and shared data will support reproducibility.

1.6 Scope and limitations

Shares were recorded at the acquisition date of 8 January 2015, giving older articles longer to accumulate shares [1]. Chronological splitting does not remove this unequal exposure or fully recreate information available at past decision dates. The study therefore provides retrospective evidence about ranking. Better shortlist performance would not establish that promotion caused additional shares or revenue. Operational use would require checking feature availability, evaluating recent articles over a common outcome window, and testing promotional effects. Editors would retain the final decision.

2. Data Preparation and Exploratory Analysis

2.1 Data Cleaning

Data scope and target definition

The original dataset contains 39644 articles and 61 original fields: 58 predictive attributes, two non-predictive fields (URL and timedelta), and the outcome shares. A binary target, popular, is derived for later classification. It equals 1 only when shares are greater than 1400. The threshold is the sample median; 1,592 articles have exactly 1400 shares, so stating the strict greater-than rule avoids ambiguity. After cleaning, 39,643 articles remain, of which 19561 (49.3%) are popular.

Raw articles	Removed	Cleaned articles	Popular rate
39644	1 invalid record	39643	49.3%

Cleaning sequence

1. Removed leading spaces from column names so that all variables use consistent names.
2. Removed one record because at least one word-ratio value fell outside the valid range from zero to one. This is mathematically invalid rather than a plausible extreme value.
3. Retained 1181 zero-length articles (2.98% of the sample) and created an empty-content flag. These records may represent video or gallery pages, or text that was not captured. Deleting them could therefore bias the content-format analysis.
4. Grouped 6134 articles (15.5%) without a named channel as Other or Unclassified. Removing this substantial group would reduce the sample and could distort comparisons among channels.
5. Excluded URL because it is an identifier rather than a generalisable predictor. The variable timedelta was reserved only for chronological ordering because it is measured after publication and reflects the article's position in the collection period.

Output of this step

Cleaning removes the single invalid record but retains unusual, potentially meaningful article formats and genuine high-share outcomes. Shares are strongly right-skewed: the median is 1400, the mean is 3395, the 95th percentile is 10800 and the maximum is 843300. For this reason, the analysis focuses on popular rates rather than the mean number of shares. Where an axis ends at the 99th percentile, the cutoff is for display only; no observations are deleted.

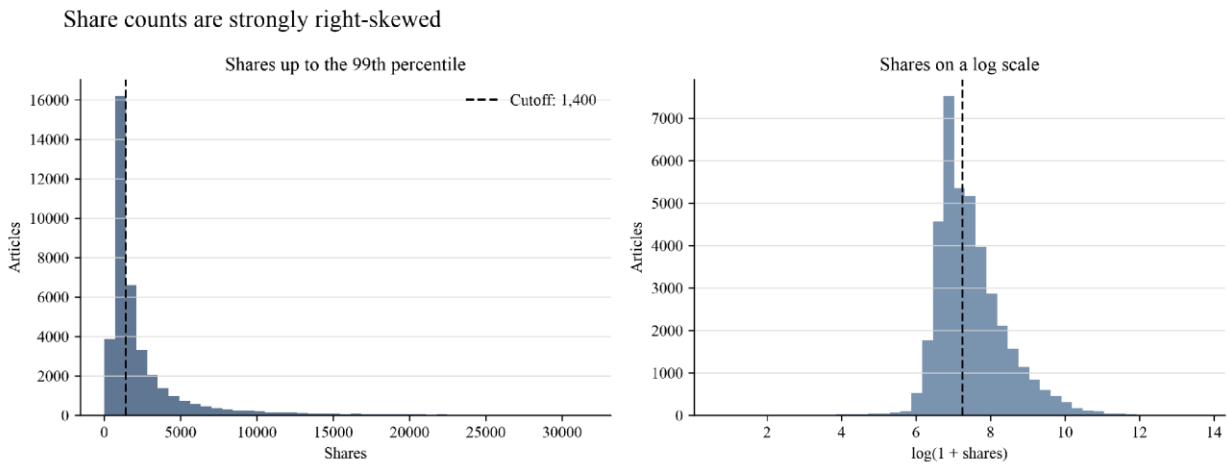


Figure 2.1 – Distribution of article shares and the popularity cutoff

2.2 Data Split

Reason for a time-based split

The intended use is to learn from earlier articles and assess performance on later articles. A random split mixes observations from different periods and may hide changes in audience behaviour or publishing practice. A chronological split therefore provides a more realistic test of future performance and prevents later observations from influencing model development. It is deliberately stricter than a random split: the two approaches answer different questions rather than one being universally correct.

Split rule

The variable timedelta records the number of days between publication and data collection, so a larger value represents an earlier article. Complete timedelta blocks were assigned in chronological order to avoid placing articles from the same collection day in different subsets. The resulting training, validation and test samples contain approximately 60%, 20% and 20% of the cleaned data.

Dataset	Timedelta range	Articles	Popular rate
Training	265 to 731	23793 (60.0%)	50.9%
Validation	132 to 264	7966 (20.1%)	50.5%
Test	8 to 131	7884 (19.9%)	43.5%

Table 2.1 – Table summarizing dataset split

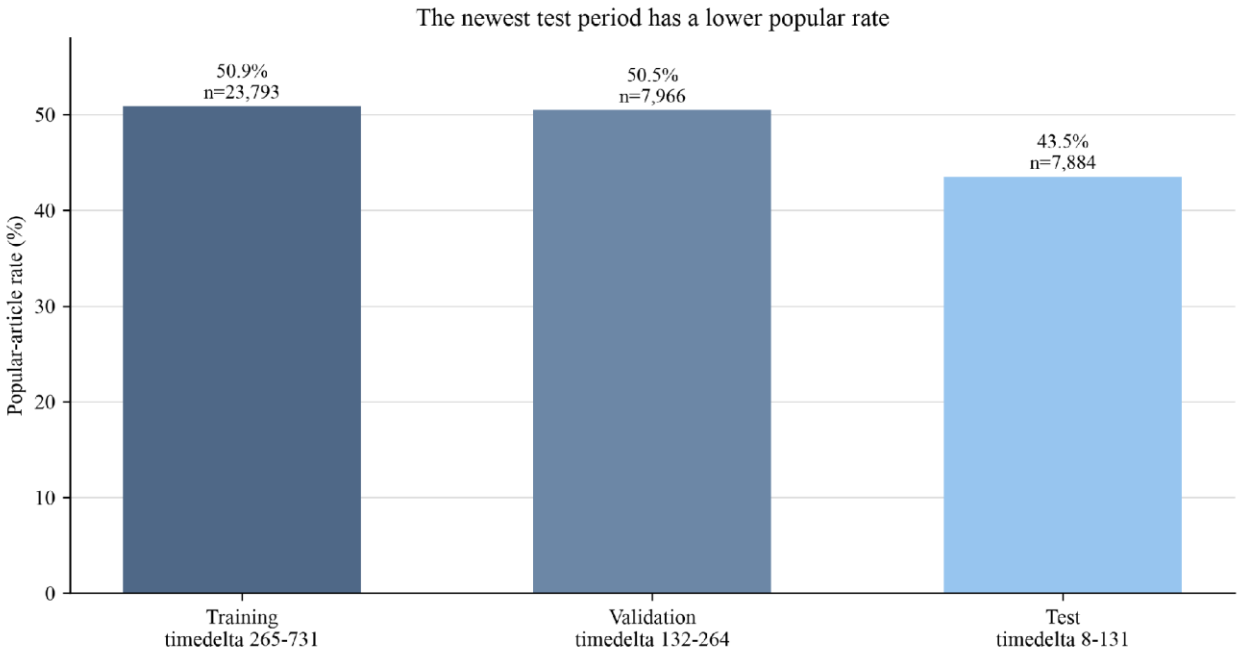


Figure 2.2 – Popular-article rate across the training, validation and test periods

Interpretation

The popular rate declines from 50.9% in training to 50.5% in validation and 43.5% in testing. The 7.4 percentage-point training-to-test difference shows target drift: popular articles are less common in the latest period. A model can therefore appear weaker on the test set partly because the later period is different, not only because of poor fitting. Model comparisons should use the same split and report it clearly. Team members may also use random splitting for benchmark models, but those results measure performance under a more stable-population assumption.

2.3 Publishing Time

Measure and comparison

The analysis uses popular rate rather than the number of popular articles. For each group, the rate equals popular articles divided by all articles in that group. This denominator is essential because the dataset contains five weekdays but only two weekend days; raw counts would mainly reflect how many opportunities each day type has to publish articles. The rate instead estimates the observed probability that an article in the group is popular.

Timing group	Articles	Popular articles	Popular rate
Weekday	34453	16080	46.7%
Weekend	5190	3481	67.1%

Table 2.2 – Table summarizing difference in timing groups

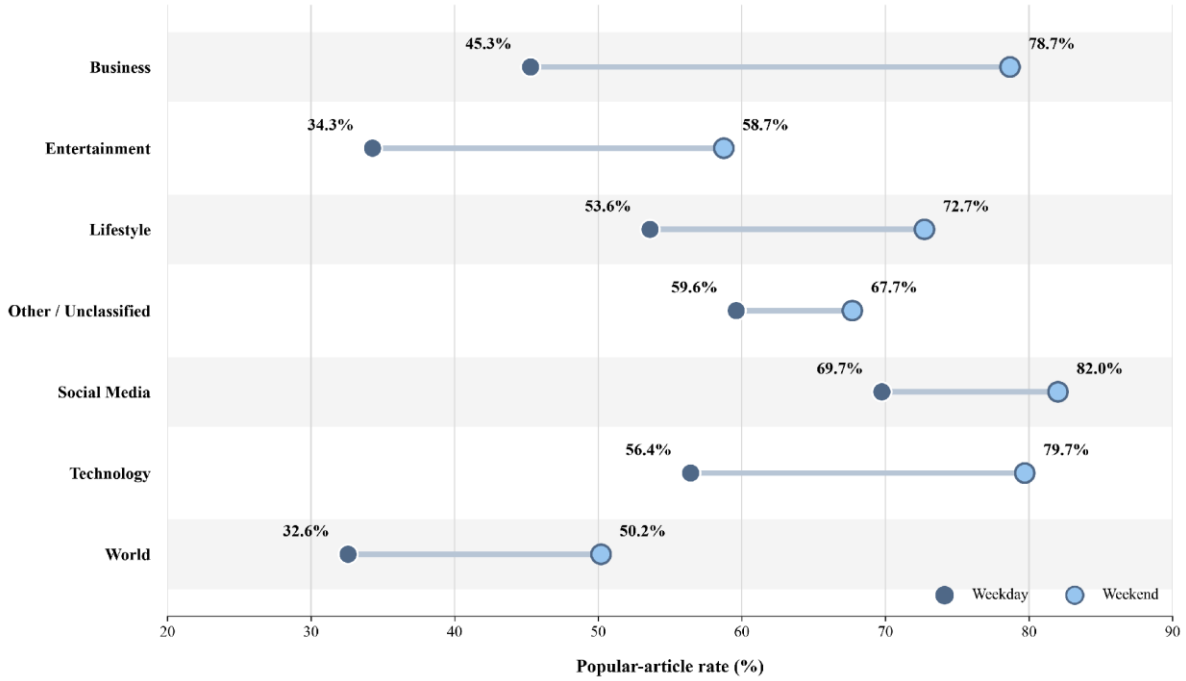


Figure 2.3 – Observed popular article rates for weekdays and weekends by channel

Interpretation

The overall weekend popular rate is 67.1%, compared with 46.7% on weekdays, a difference of 20.4 percentage points. Every channel shows the same direction, but the size varies. Business rises from 45.3% to 78.7% (+33.4 points), while Other or Unclassified rises from 59.6% to 67.7% (+8.1 points). Entertainment and Technology also show differences above 20 points. This variation suggests that weekend timing and channel may interact in later models. These are unadjusted associations: topic selection, editorial scheduling and audience behaviour may explain part of the pattern, so the chart does not establish that weekend publication causes popularity.

2.4 Content Format

Analysis method

This step examines article length, image count and video count using observed popular rates without a fitted curve. Positive article lengths are divided into 20 equal-size groups and plotted at the group median; zero-length articles form a separate category. Image and video counts are shown at observed integer values. Point size represents sample size, vertical lines show 95% Wilson confidence intervals, and the dashed line marks the 49.3% overall rate. Axes end at the 99th percentile only for display; no records are removed.

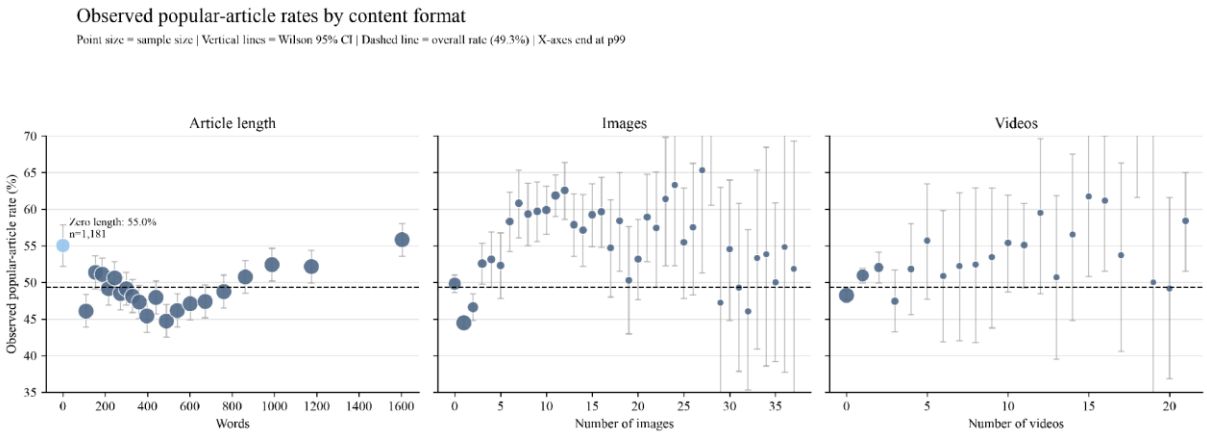


Figure 2.4 – Observed popular-article rates by article length, image count and video count

Observed patterns

Article length: The observed relationship is U-shaped. The popular rate falls toward approximately 400 to 500 words and then rises for longer articles, while zero-length articles have a rate of about 55%. This pattern supports allowing a non-linear effect in modelling, but it does not identify an optimal article length because the groups also differ in topic, channel and other content features.

Images: Popular rates generally rise across lower image counts, but the high-count estimates fluctuate and their confidence intervals widen as the number of articles falls. The figure therefore supports a possible non-linear association; it does not support the rule that adding more images will always increase popularity.

Videos: The relationship is weaker and has no stable increasing or decreasing pattern. Most articles contain few videos, so estimates in the high-count tail are based on small samples and have wide confidence intervals. The available evidence is not strong enough to claim that video count consistently helps or harms popularity.

Use in the next stage and limitations

Later models should keep the zero-length flag, allow non-linear effects for article length and image count, and test weekend and channel together. These features should be evaluated jointly because the descriptive patterns may change after controlling for topic, channel and other article characteristics.

Limitations: The dataset records the day of the week but not the publication hour or a complete calendar date, so more detailed timing effects cannot be tested. Shares are cumulative, and articles may have different amounts of time or exposure to accumulate them. All findings in this section should therefore be read as descriptive evidence and modelling hypotheses, not as causal or universal editorial rules.

3. Logistic Regression and Adjusted Associations

3.1 Model Setup

Purpose and outcome

This section asks which article characteristics remain associated with popularity when the other measured factors are considered together. We fit one ordinary logistic regression using all eligible predictors. Keyword statistics, publication weekday and channel show the strongest

statistical evidence. For prediction, the main result is a precision of 61.95% at a 0.5 cutoff: about 62 out of every 100 articles predicted to be popular were actually popular.

The outcome follows Part 2: popular equals 1 when shares are greater than 1,400 and equals 0 otherwise. We retain the chronological split and fit the model on the 23,793 training articles only. The validation and test sets contain 7,966 and 7,884 articles. The reported prediction results use the test set and the standard 0.5 cutoff.

Predictors included together

All 67 cleaned-data fields were reviewed. The final model contains 45 predictor fields, which expand to 55 coefficients plus an intercept after category coding. The table shows the coverage of the model. The six factors discussed later are selected for explanation only; they do not replace the full set of predictors.

Predictor group	Number of fields
Sentiment and subjectivity	16
Keywords and their statistics	10
Length and word use, including empty content	6
Media and links	4
Topic proportions	4
Referenced-article share statistics	3
Channel and publication weekday	2
Total	45

Table 3.1 – List of predictor groups and respective number of fields

Coding and fitting

Shares are excluded from the predictors because it defines the answer. URL is an identifier, and timedelta and temporal_split are used for data organisation only. Channel and weekday replace their duplicate indicator columns; weekend and period are already represented by weekday. One topic proportion is the reference because the five proportions sum to one. The nearly duplicate n_non_stop_words field is omitted while the empty-content flag is retained.

Numeric predictors are centred and scaled using training-set means and standard deviations. The same values are applied to later articles. The model uses a binomial response with a logit link, fitted in R with glm. It contains main effects only, with no splines, interactions or stepwise variable selection. This gives one adjusted coefficient for each numeric predictor, rather than a separate curve over its range.

Checks before interpretation

The fitting procedure converged in five iterations. Collinearity checks found nine model columns with VIF above 10, with a maximum of 320.5. This means that some predictors overlap strongly, so individual coefficients need care. Influence screening flagged 504 training observations using Cook’s distance above 4/n. These are review flags, not proof of bad data; no observations were removed on that basis.

3.2 Factors with the Strongest Statistical Evidence

Comparison across all factors

We compare each of the 45 factors by removing it from the full model and checking how much the fit changes. This likelihood-ratio test asks whether the factor adds information after the other included variables have been accounted for. Weekday and channel are tested as complete factors, with all their category coefficients considered together.

Table 3.1 lists the six smallest overall p-values. These are the strongest statistical signals in this fitted model. A smaller p-value does not necessarily mean a larger association or a better predictor on its own. We also apply the Benjamini–Hochberg adjustment across the 45 tests; all six remain significant at the 5% level.

Factor	CSV field	Overall p-value
Keyword mean-share statistic	kw_avg_avg	9.86e-89
Publication weekday	weekday	1.75e-78
Article channel	channel	7.25e-66
Keyword max-share statistic	kw_max_avg	1.25e-41
Absolute title subjectivity	abs_title_subjectivity	2.82e-08
Content length	n_tokens_content	3.66e-08

Table 3.2 – Six leading factors from the full-model likelihood-ratio tests

What this comparison tells us

The two keyword statistics, weekday and channel stand out clearly. They suggest that the context of an article matters alongside its wording and length. Absolute title subjectivity and content length also retain evidence of an association after adjustment, although their p-values are less extreme than those of the first four factors.

The keyword labels refer to the dataset fields kw_avg_avg and kw_max_avg. They describe different summaries of keyword share statistics. They should not be read as instructions to choose a particular keyword. Weekday and channel each contain several categories, so their overall p-values establish category differences without giving a single direction for the whole factor.

This comparison covers the full predictor set. It does not assume that length, images and videos are the only relevant factors, and it does not drop the remaining predictors simply because they are absent from the table.

3.3 Direction and Size of the Associations

How to read an adjusted odds ratio

An odds ratio (OR) above 1 indicates a positive association with popularity; an OR below 1 indicates a negative association. Each numeric OR below refers to an increase of one training-set standard deviation, with the other predictors held constant. This common scale helps compare variables measured in different units. An OR of 2.60 means 2.60 times the odds, not 2.60 times the probability.

Numeric factor	Adjusted OR	95% confidence interval

Keyword mean-share statistic	2.60	2.36 to 2.87
Keyword max-share statistic	0.56	0.52 to 0.61
Absolute title subjectivity	1.10	1.06 to 1.13
Content length	1.15	1.10 to 1.22

Table 3.3 – Numeric factors and respective adjusted OR and 95% confidence interval

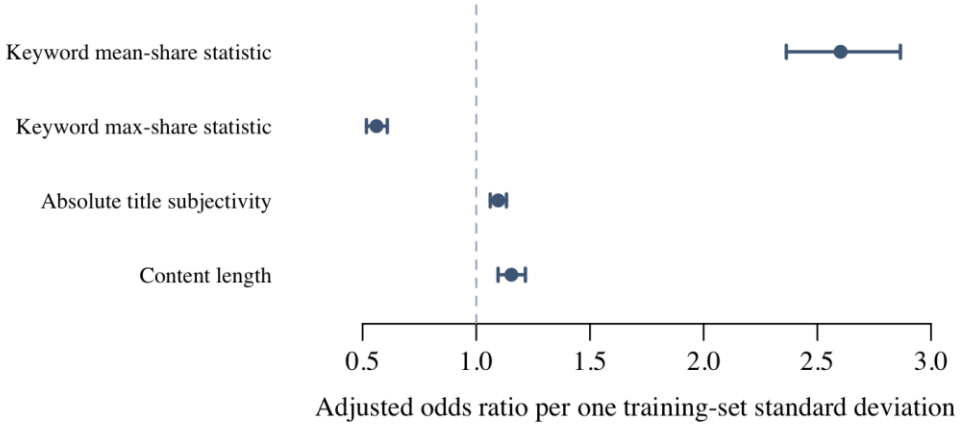


Figure 3.1 – Adjusted associations for the four numeric factors among the leading six

Reading the results

The keyword mean-share statistic has the largest positive OR among these four variables: 2.60, with a 95% confidence interval from 2.36 to 2.87. The keyword max-share statistic has an OR of 0.56, with an interval from 0.52 to 0.61. These opposite signs are conditional on the other predictors, including related keyword statistics. They do not show that using more popular keywords reduces an article's chance of success.

Absolute title subjectivity has a smaller positive association, with an OR of 1.10. Content length has an OR of 1.15. All four intervals lie entirely on one side of 1. The length result describes the average linear association on the log-odds scale used in this model. It does not establish an ideal article length or reproduce the curved pattern seen in the exploratory analysis.

Weekday and channel comparisons

Categorical factors need a reference category rather than a single overall OR. With Monday as the weekday reference, Saturday has an adjusted OR of about 2.41 and Sunday about 1.80. Relative to Other or Unclassified, social media has an OR of about 2.08 and Technology about 1.45. These comparisons come from the same full model; they are associations, not evidence that changing the publication day or channel alone would cause an article to become popular.

3.4 Prediction of Popular Articles

The 0.5 classification cutoff

The model assigns each test article a probability of being popular. A probability of at least 0.5 gives a predicted label of popular; a lower probability gives a predicted label of not popular. The true label still depends on whether shares exceed 1,400. The matrix below expresses each outcome as a share of all 7,884 test articles, so its four inner cells add to 100%.

Actual / Predicted	Not popular	Popular	Total
Not popular	41.12%	15.35%	56.47%
Popular	18.54%	24.99%	43.53%
Total	59.67%	40.33%	100%

Table 3.4 – Test confusion matrix at a 0.5 cutoff as percentages of all test articles

Precision for the predicted popular class

Of the 3,180 articles predicted to be popular, 1,970 were actually popular and 1,210 were not. Precision uses only these predicted-popular articles as its denominator. It therefore answers a practical question: when the model says an article will be popular, how often is that label correct?

$$\text{Precision} = 1{,}970 / (1{,}970 + 1{,}210) = 61.95\%$$

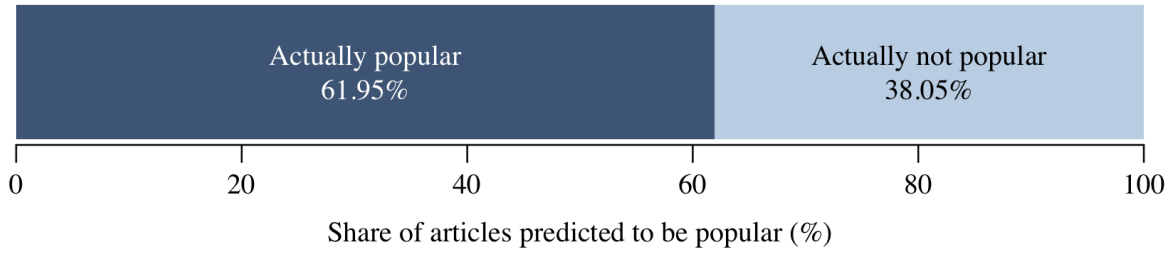


Figure 3.2 – Actual outcomes among articles predicted to be popular

In plain terms, about 62 in every 100 predicted-popular articles were genuinely popular, while about 38 were false positives. This rate is higher than the 43.53% popular rate across the whole test set. It shows that the positive predictions contain useful information, but they are far from certain.

Precision and accuracy measure different things

Precision is not the percentage of all articles classified correctly. Overall accuracy includes correct non-popular predictions as well: (3,242 + 1,970) / 7,884 = 66.11%. The model also misses 1,462 actually popular articles. For that reason, the 61.95% precision result should not be presented as finding 61.95% of all popular articles.

Scope of the result

The full model provides adjusted comparisons across the available factors and a probability for each article. The strongest statistical signals come from keyword statistics, weekday and channel. The 61.95% precision result describes the positive predictions at the standard 0.5 cutoff in this particular test period; it is not a promise for future articles.

Earlier exploratory work examined other cutoffs on these same test data. The figures here return to 0.5 and should not be described as a fresh, untouched test. Historical keyword and referenced-article share statistics also need to be available before publication for a real prediction system. The CSV does not establish when those aggregates were calculated.

4. CART Decision Tree Model

4.1 Rationale for adopting the CART classifier

This project aims to identify which news articles will become popular. Many explanatory features are available in the cleaned dataset, yet it is not straightforward to translate raw variables into practical prediction rules. Analysts face several common difficulties: determining meaningful cutoff thresholds for continuous metrics, assigning reasonable weights when multiple conflicting indicators exist, and selecting effective feature combinations without triggering overfitting.

A Classification and Regression Tree (CART) offers an interpretable, data-driven solution to these challenges. Instead of manually defining decision rules, the CART algorithm learns binary splitting rules directly from training data. It automatically searches for optimal thresholds and selects feature combinations that maximise class separation for the target variable.

The target label popular is binary: articles are coded as either *Popular* or *NotPopular*. We performed time-based dataset partitioning to avoid data leakage, producing three independent subsets:

- Training set: 23 793 observations for growing the initial tree
- Validation set: 7 966 observations for hyper-parameter tuning and pruning
- Held-out test set: 7 884 observations for final out-of-sample evaluation

In total, 48 candidate explanatory features were supplied to the model. These include content metadata, keyword share statistics, topic outputs from latent Dirichlet allocation (LDA), sentiment metrics, and publication-time related variables.

4.2 Model training, cost-complexity pruning and feature filtering

We first trained a full-depth classification tree on the training sample. An unpruned full tree tends to overfit: it memorizes random noise and idiosyncratic patterns within training observations, which degrades performance on unseen data.

Cost-complexity pruning addresses overfitting by introducing the complexity-parameter (CP). Larger CP values remove more tree branches. Branches that deliver little improvement in validation-set classification performance are discarded. We iteratively pruned the full tree across a range of CP candidates and computed misclassification error on the validation dataset for each pruned tree. The optimal CP value was selected at the point which minimises validation-set error, giving bestCP = 0.000813.

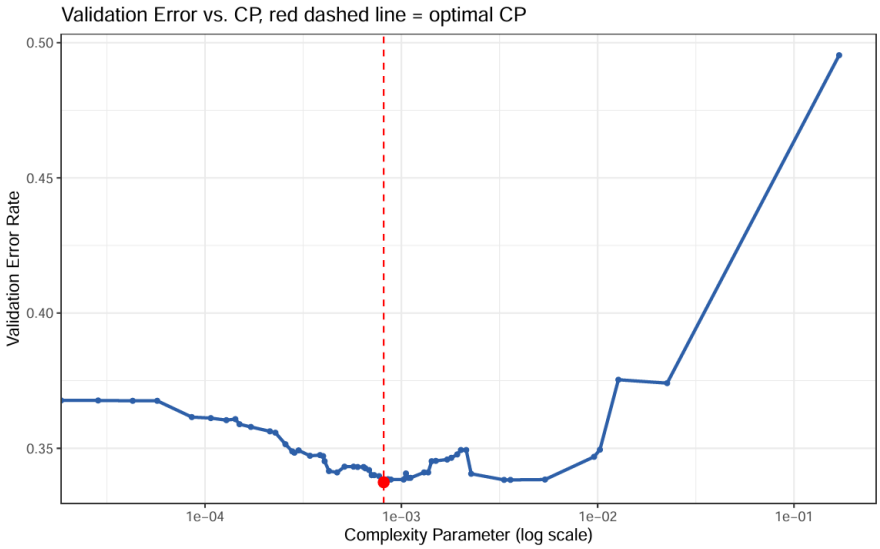


Figure 4.1 – Validation Error Rate against Complexity Parameter (CP) for CART Pruning

After pruning, only **25 out of the original 48 candidate features** were used for tree splitting. Many input variables carried limited predictive signals and were entirely excluded from the final tree structure. This demonstrates the pruning process acts as an embedded feature-selection step, stripping away redundant predictors.

4.3 Feature importance analysis

Variable importance for the CART model quantifies the total reduction in Gini impurity produced by each feature across all tree splits. It reflects how strongly each predictor helps separate popular articles from non-popular ones. After cost-complexity pruning, the final tree only retains 25 features. All these 25 variables actively participate in tree splitting and contribute to prediction; features with no predictive value are fully discarded during pruning and do not appear in the final model.

Ranked by variable importance, the top five predictors are:

1. **Channel:** News article category, the most influential predictor
2. **kw_avg_avg:** Average share count of keywords associated with the article
3. **LDA_04:** One latent topic dimension generated by LDA topic modelling
4. **kw_max_avg:** Maximum share value among keywords of the article
5. **is_weekend:** Binary indicator showing whether the article was published on Saturday or Sunday

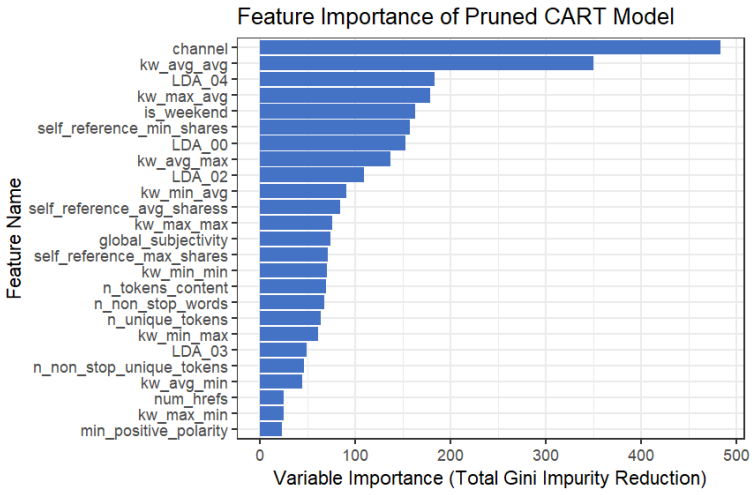


Figure 4.2 – Feature Importance of Pruned CART Model

The remaining 20 retained predictors include self-reference share statistics, other LDA latent topic dimensions, additional keyword share metrics, content length measurements, and text sentiment indicators. While their importance values are lower than the top five features, all of them contribute to at least one split in the pruned decision tree and support the classification task.

4.4 Out-of-sample performance on the test set

Our core objective is successfully detecting genuinely popular articles. For this use case, **precision carries higher practical priority than overall accuracy**. Precision answers the business-relevant question: among all articles predicted as Popular, what proportion are truly popular.

A random-guess baseline was established via 1000 simulation runs. Random predictions follow the natural proportion of popular articles within the test set (43.53%). The simulated random baseline yields an average overall accuracy of 50.78%.

For the CART model, we determined the optimal probability threshold 0.45 from the validation set by maximising Youden's J statistic (sensitivity + specificity − 1). Applying this threshold to the held-out test set produces the following results:

- Test-set overall accuracy: 65.68%, representing a ~14.9 percentage point improvement over random guessing
- Test-set precision for the *Popular* class: 50.8%
- Validation AUC: 0.6953; Test-set AUC: 0.6822

The test-set confusion matrix is shown below:

Actual / Predicted	Popular	Not Popular
Popular	1,909	1,523
Not Popular	1,183	3,269

Table 4.1 – Final CART Model Confusion Matrix

Total correct predictions equal 5178 out of 7884 test samples. While classification performance is far from perfect, the CART model delivers meaningful predictive power beyond uninformed random guesses.

4.5 Section conclusion

The CART decision tree builds interpretable classification rules for article popularity without manual definition of thresholds or feature weights. Cost-complexity pruning effectively mitigates overfitting and automatically filters low-value features, shrinking 48 input variables down to 25 practically used predictors.

Article category, keyword-share metrics, latent content topics and weekend publication status dominate prediction outcomes. Evaluated on completely unseen test data, the model outperforms random baseline performance. Most critically, it achieves reasonable precision for identifying popular articles, which is the primary goal of this modelling exercise. Although there remains space for further performance improvement, CART provides a transparent baseline solution for this classification task.

5. Random Forest Model

5.1 Purpose and Goal

Following the successful implementation of the CART classification model, which improved prediction accuracy from the baseline of 43.5% to 61.7%, the potential for further improvements in predictive performance was explored through the application of a more complex machine learning algorithm. One such approach is the Random Forest model, which, unlike CART, constructs multiple decision trees rather than a single tree. The predictions generated by these trees are subsequently aggregated through majority voting to produce a final classification, potentially improving predictive accuracy and model robustness.

Similar to the CART model, the Random Forest model aims to predict whether an online news article will be classified as popular or not popular using 47 predictor variables. Should the Random Forest model demonstrate superior predictive accuracy compared to the CART model, it could be adopted by Mashable's business analytics team to predict the popularity of future articles and rank them according to their estimated probability of being popular.

5.2 Methodology

RStudio was employed as the primary development environment for the implementation, training, and evaluation of the Random Forest classification model using the R programming language. The model was initialized using the randomForest package, while the data.table and ggplot2 packages were imported to support data manipulation and visualisation.

Additional Data Cleaning

To ensure that the dataset was suitable for Random Forest modelling, additional data cleaning steps were taken. Variables considered irrelevant, redundant, or potentially responsible for data leakage were excluded from the modelling process. Consequently, each of the 3 datasets—training, validation, and testing—comprised of 47 predictor variables and 1 target variable. An additional copy of the testing dataset was retained with the article URL variable to facilitate the identification and labelling of individual articles.

Model Training and Hyperparameter Optimization

The Random Forest model was initially trained using the training dataset. Two hyperparameters were selected for optimisation: the number of trees (B) and the number of randomly selected features (RSF) considered at each decision-tree split.

A grid search was conducted using three values for each hyperparameter, resulting in a total of nine distinct hyperparameter combinations:

- Number of trees (B): 100, 300, and 500.
- Randomly selected features (RSF): 3, 6, and 12.

Each model was fitted to the training dataset and subsequently evaluated using the validation dataset. Model performance was assessed using classification accuracy, false positive rate (FPR), and false negative rate (FNR), calculated from the corresponding confusion matrix.

Classification accuracy measures the proportion of articles correctly classified as either popular or not popular. The FPR measures the proportion of actually unpopular articles incorrectly classified as popular, while the FNR measures the proportion of actually popular articles incorrectly classified as unpopular.

Additionally, the out-of-bag (OOB) error was recorded for each model. OOB error provides an internal estimate of classification error using observations excluded from the bootstrap sample used to construct each tree.

The performance of each model is summarised in table 5.1 as shown below.

RSF	B	OOB Error	Accuracy	FPR	FNR
3	100	36.35%	66.86%	32.77%	33.51%
3	300	34.99%	67.20%	33.27%	32.34%
3	500	34.65%	67.39%	32.77%	32.46%
6	100	35.75%	67.06%	32.11%	33.76%
6	**300**	**34.71%**	**67.90%**	**31.93%**	**32.26%**
6	500	34.51%	67.71%	32.03%	32.54%
12	100	35.96%	66.46%	32.72%	34.35%
12	300	34.70%	67.36%	32.79%	32.49%
12	500	34.48%	67.31%	33.05%	32.34%

Table 5.1 – Performance Summary for each hyperparameter combination model

Classification accuracy was selected as the primary optimisation criterion. The combination of an RSF value of 6 and a B value of 300 was identified as the optimal model configuration, achieving the highest validation accuracy while also recording the lowest FPR and FNR.

Final Model Development and Evaluation

Following hyperparameter optimisation, the training and validation datasets were combined to maximise the number of observations available for final model development. The Random Forest model was then retrained using the selected optimal values of B and RSF.

The final model was subsequently evaluated using the testing dataset. The predicted article popularity classifications were compared against the actual classifications to construct a confusion matrix.

The model's final classification accuracy, FPR, and FNR were calculated to assess predictive performance on previously unseen observations. The final OOB error was also recorded, and the Random Forest error curve was generated to examine how classification error varied with the number of trees.

Predictor Importance Analysis

To identify the variables contributing most strongly to classification performance, a predictor importance analysis was conducted. Variable importance was assessed using the Mean Decrease in Accuracy measure, which evaluates the reduction in predictive accuracy when

the values of a particular predictor are randomly permuted among OOB observations. The resulting variable importance plot was used to identify and rank the most influential predictors.

Popularity Probability Estimation

In addition to binary classification, the Random Forest methodology was used to estimate the probability of each article being classified as popular. These probabilities represent the proportion of decision trees voting for each classification outcome.

The predicted probabilities were subsequently combined with the corresponding article records, including their URLs and actual popularity classifications. Articles were then ranked in descending order according to their predicted probability of being popular.

5.3 Results and Analysis

Model Performance and Classification Accuracy

The final model achieved a classification accuracy of 67.45% and a precision of approximately 62.26%, demonstrating its ability to predict the popularity of online news articles. The confusion matrix summarising the model's predictions against the actual classifications is presented below.

Actual / Predicted	Popular	Not Popular
Popular	2,199	1,233
Not Popular	1,333	3,119

Table 5.2 – Final Random Forest Model Confusion Matrix

From the confusion matrix, the resulting evaluation metrics are summarised below.

Evaluation Metrics	Result
Classification Accuracy	67.45%
Precision (Popular)	62.26%
True Positive Rate (TPR)	64.07%
True Negative Rate (TNR)	70.06%
False Positive Rate (FPR)	29.94%
False Negative Rate (FNR)	35.93%

Compared to the previously developed CART model, which achieved a classification accuracy of 61.7%, the Random Forest model demonstrated an improvement of 5.75 percentage points. This improvement suggests that aggregating predictions from multiple decision trees can enhance predictive performance compared to relying on a single decision tree.

The improvement in classification accuracy achieved by the Random Forest model demonstrates its potential as a decision-support tool for Mashable's business analytics team. Compared to the CART model, the Random Forest model provides a more accurate approach to predicting whether an online news article is likely to be popular.

Additional model refinement and evaluation may further improve predictive performance, particularly in reducing false negative classifications and validating the reliability of predicted popularity probabilities.

Out-of-Bag Error Analysis

The Random Forest error curve illustrates the relationship between the number of decision trees and the model's out-of-bag (OOB) classification error.

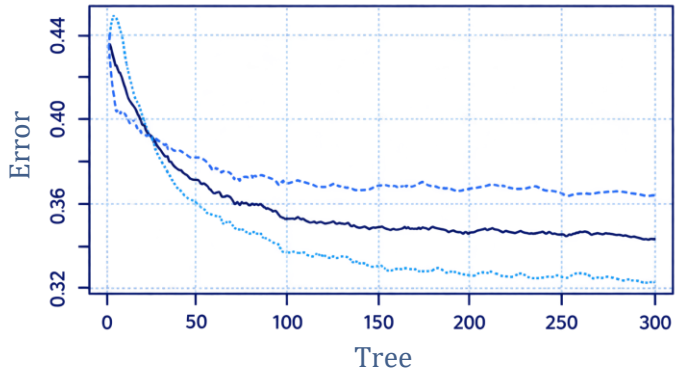


Figure 5.1 – Final Model Out-of-Bag (OOB) Error Plot

As shown in the Random Forrest error curve, beyond approximately 150–200 trees, the rate of improvement became progressively smaller, with the overall OOB error stabilising at approximately 34% as the model approached 300 trees. Overall, the error curve supports the selection of 300 decision trees as an appropriate configuration, achieving relatively stable classification performance, with additional trees providing increasingly limited improvements in classification performance.

Predictor Variable Importance Analysis

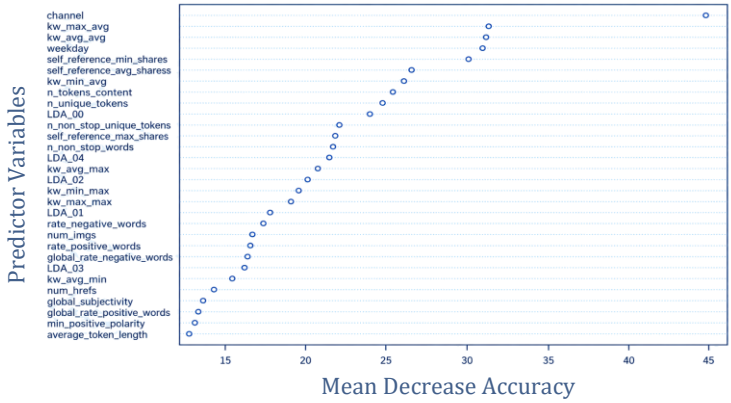


Figure 5.2 – Final Model Variable Importance Plot

Based on the variable importance plot, the five most influential predictors were identified as follows:

1. **Channel:** The most influential predictor, indicating that the subject matter of an article is strongly associated with the popularity and differences in audience interest across categories such as technology, entertainment, and business may contribute to variations in article popularity.
2. **Maximum Average Keyword Shares:** Its relatively high importance suggests that keywords previously associated with highly shared articles provide useful information for predicting future popularity.
3. **Average Keyword Shares:** This indicates that the broader historical popularity of an article's keywords may also contribute to predicting its likelihood of being shared.

4. **Publication Weekday:** The importance of this variable suggests that variations in audience browsing behaviour and online engagement throughout the week strongly affect an article's popularity.
5. **Minimum Shares of Referenced Articles:** Its predictive importance suggests that the historical popularity of related content may provide useful information about an article's potential performance.

These findings provide useful insights into the characteristics associated with the most popular online news articles that can be passed on to writers and editors to create more appealing news articles in the future.

6. Evaluation & Recommendation

6.1 What we were trying to answer

Imagine we are Mashable's data team. The company wants to push articles through paid promotion — homepage slots, social media ads, newsletter features — to get more clicks, more shares and hopefully one or two breakout stories. Breakout stories mean ad revenue and brand visibility. But the promotion budget is small: out of roughly 15 new articles a day, editors only have slots for three. The rest go out normally and live or die on their own momentum.

So, the question for us is simple: can we look at an article the moment it goes live and tell which ones are actually going to be popular, so the three promotion slots go to the right stories? We defined a popular article as one with more than 1,400 shares, and trained three different models on roughly 32,000 historical Mashable articles from 2013–2014. We then compared them on a held-out test set of 7,884 articles, split by time so the test set is the newest period. Precision is the main metric, because promoting a non-popular article wastes real money — far more costly than missing a potentially popular one.

We tested Logistic Regression (which tells us which features matter and by how much), a CART decision tree (which gives editors readable if-then rules), and a Random Forest (300 trees, which tests whether a more flexible model does better). All three were evaluated on the same articles and under the same 20% capacity rule.

6.2 Which model wins

Random Forest came out on top on both precision and accuracy, though not by a huge margin.

Model	Precision	Accuracy
Logistic Regression	61.95%	66.15%
CART	61.70%	65.70%
Random Forest	62.25%	67.45%

Table 6.1 – Summary of predictive model performances

The gap between models is genuinely small. Random Forest beats Logistic Regression by only 0.3 percentage points in precision, and by 1.3 points in accuracy. CART is a touch behind on both. Against a 50.8% coin-flip baseline, Random Forest gets about 17 points more right overall. The out-of-bag error on the training side was 33.89%, which is consistent with the test-set accuracy and suggests the model isn't badly overfitting.

It's not a landslide — but it's consistently ahead on both metrics, and on a test set of nearly 8,000 articles a 0.3-point precision edge still means about 24 more correctly flagged winners. That's enough for us to recommend Random Forest.

6.3 Does it actually help editors?

Precision at a 0.5 cutoff isn't really what editors use. In practice they take a ranked shortlist and look at the top stories, so that's the number that matters for the budget decision. When we scored every test article with Random Forest and took the predicted top 20% (1,577 articles), 599 of them turned out to be popular.

Shortlist strategy (top 20%)	Hit rate
Picking at random	20.0%
Random Forest shortlist	37.98% (1.90× lift)

Table 6.2 – Shortlist strategy and respective hit rate

That's a 37.98% hit rate — nearly double the 20% you'd get by picking blind. It means that for every 10 articles editors would have pushed anyway, about two more of them actually go popular. The promotion budget isn't doubled, but it's spent roughly twice as well. To put it another way: of the 1,577 articles in the shortlist, 978 still aren't popular, so the model doesn't magically solve the problem — but it shifts the odds clearly in editors' favour.

6.4 Where this score can be used

The model takes all 58 features and outputs one popularity score for each new article. Based on what the feature importance shows — channel, keyword statistics and topic proportions dominate — we see four practical places this score is useful:

- **Homepage ranking.** Score every new article and use the rank to decide which headlines earn the top editorial slots. Editors still have the final call, but the score gives them a defensible starting order.
- **Paid promotion budget.** Shift marketing spend towards high-score articles so each promotion dollar buys more shares. This is where the 1.90× lift directly translates into money.
- **Topic and pitch planning.** Feature importance says channel, keywords and topic proportions drive popularity most. Use those signals to guide what the newsroom pitches next week — for example, pushing more stories into channels and topic clusters that historically over-perform.
- **Publishing timing.** Weekend articles share at about 67% popular rate versus 47% on weekdays, and the gap holds across all channels. Use that to suggest the best day and slot for a given story, rather than just dumping everything on Monday morning.

6.5 Limitations

Before treating this as a finished product, there are four caveats worth flagging:

- **The data is old and narrow.** All articles are from Mashable in 2013–2014, and the popular rate already dropped from about 51% in training to 43.5% in the newest test period — a 7.4 percentage-point decline. Reader habits, social platforms and content formats have moved on since then. The model will need regular retraining on fresh data, and it shouldn't be copied to other publishers or other languages without re-checking.
- **The three models are really close.** Random Forest only beats Logistic Regression by 0.3 points in precision. If explainability or simplicity matters more than that tiny edge — for example, if editors want to read off coefficients and trust the model — the simpler Logistic Regression may actually be the better choice in production.
- **We only use what we know before publication.** The model looks at article features the moment a story goes live — length, keywords, sentiment, channel, media count,

referenced-article shares — but it ignores early clicks and first-hour social momentum. Those real-time signals usually predict virality better, and a live system should combine this pre-publication score with early engagement data rather than rely on it alone.

- **It's a support tool, not a verdict.** Even with the best model, around 62% of the top-20% shortlist is not actually popular, and the 1,400-share cutoff is just our project convention — a different threshold would move all the numbers. Editors should still use their own judgment; the score narrows the list, it doesn't replace the editor.

6.6 Conclusion

Random Forest is the best of the three we tested: 62.25% precision, 67.45% accuracy, and a 1.90× lift when we shortlist the top 20% of articles. We recommend shipping it as a decision-support score for homepage ranking, paid promotion, topic planning and publishing timing — with the caveat that it needs regular retraining on newer data, the threshold should be re-calibrated to what the business actually considers a breakout story, and it should never override an editor's judgment on which story deserves to be told.

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